Miriyala Dhanush Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.04
sri chaitanya	Class XII (ssc)	2024	98.00%
sri chaitanya	Class X (ssc)	2022	10.00%

PROJECTS

1. Credit Card Approval -- Independent Credit Model Validation

- Developed a credit-card approval classification framework using **Logistic Regression, Gradient Boosting/XGBoost, and Random Forest**, benchmarking a champion model against challenger models.
- Independently validated model performance using **ROC-AUC, KS, Gini, precision, and recall**, evaluating discrimination, classification trade-offs, and performance deterioration across validation samples.
- Performed **out-of-time validation, feature-stability analysis, and Population Stability Index (PSI)** testing to assess model robustness under changes in applicant populations and identify potential model drift.
- Audited model inputs for **data leakage, missing values, outliers, and sampling issues**, and documented validation findings, model limitations, and recommended remediation actions.

2. NASA Engine Predictive Health -- Independent Validation & Model Risk Framework

- Developed an **XGBoost champion model** for turbofan engine health prediction on NASA C-MAPSS data and benchmarked performance against **Random Forest and baseline challenger models**.
- Conducted independent validation using **time-based out-of-time splits** to assess generalization to previously unseen engines and quantify performance degradation outside the development sample.
- Audited the modelling pipeline for **data leakage, feature instability, and temporal drift**, measuring changes in sensor distributions and their relationship with deterioration in model performance.
- Applied **SHAP-based analysis** to assess whether influential model drivers remained consistent between development and future data, and consolidated performance, stability, leakage, and risk findings into a structured **model validation report**.

3. Sales Forecasting & Demand Analytics

- Benchmarked **SARIMA, Prophet, and XGBoost** forecasting models, selecting XGBoost after reducing forecasting error by **28-29% MAE** and achieving **14% MAPE**.
- Built a four-page **Streamlit analytics dashboard** integrating forecasting, anomaly detection, and demand segmentation to support model interpretation and business decision-making.
- Detected **11 abnormal demand weeks** using Isolation Forest and cross-validated the findings against a statistical Z-score approach to independently assess anomaly signals.
- Segmented products into **four demand clusters** using K-Means, translating model outputs into interpretable demand and inventory patterns.

SKILLS

Programming Languages: Python, C, SQL

Machine Learning: Regression, Classification, Time Series Forecasting, Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA), Model Training, Model Evaluation

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, Statsmodels

Data Analytics: Data Cleaning, Data Visualization, Statistical Analysis, Business Insights

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, MySQL

POSITIONS OF RESPONSIBILITY

Overall Head | Torque, Mechanical Engineering Club, IIT Hyderabad -- *2026-Present*

- Leading an 8-member core team, overseeing technical projects, CAD initiatives, workshops, competitions, and club events while coordinating project execution and planning activities to enhance student participation.

- **Driving the club's technical initiatives and long-term planning**, organizing hands-on mechanical engineering activities and creating opportunities for students to develop practical design and engineering skills.

Sponsorship Team Member | Milan, IIT Hyderabad -- *2025-2026*

- Contributed to sponsorship outreach and coordination for IIT Hyderabad's annual inter-hostel fest, engaging with prospective sponsors, supporting partnership efforts, and assisting in building industry collaborations.

Departmental Representative | Diesta, IIT Hyderabad -- *2025-2026*

- Represented the Mechanical Engineering department, facilitating communication between students and the organizing team while promoting event participation, coordinating outreach, and supporting departmental engagement.

RELEVANT COURSES

Mathematics: Calculus, Linear Algebra, Introduction to Probability, Differential Equations, Complex Variables, Transform Techniques, Statistics, Introduction to Programming

SCHOLASTIC ACHIEVEMENTS

Maintained a **CGPA of 9.05** in B.Tech Mechanical Engineering at IIT Hyderabad.

Ranked among the **top students in a batch of 63** in the Department of Mechanical Engineering at IIT Hyderabad.

Won multiple medals across various Olympiad examinations conducted by the **Indian National Talent Search Olympiad (INTSO)**.